

# Gio Jung

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[Portfolio](#)

## Education

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### Master of Science in Computer Science

2024 - 2025

San Francisco State University

*Dissertation: [Evaluation of Vision Language Model using Item Response Theory](#)*

Advisor: Dr. Ilmi Yoon

### Bachelor of Arts in Mathematics: Concentration in Mathematics for Advanced Study

San Francisco State University

2016 - 2023

Minor in Computer Science

## Research Experience

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### YouDescribe Lab

Summer 2025 - Present

*Crowd-sourced Platform for volunteer-created audio descriptions of online videos*

- Collaborating with an interdisciplinary team to conduct research focused on improving accessibility for blind and low vision (BLV) users and communities.
- Designing and experimenting using Item Response Theory (IRT) to validate AI/LLM performances on video audio description task.
- Preprocessed and structured multi-source datasets using pandas/CSV and generated large-scale model outputs for downstream statistical and psychometric analysis.

### NAIRR Pilot Resource

Oct 2024 - Dec 2025

*“Scaling AI Video Description Model Capacity for the Blind and Low Vision Community”*

- Supported training of a large-scale multimodal video-to-text model (TimeSformer encoder  $\times$  GPT-2 decoder) on a dataset of 60K+ samples (8-frame clips, 10 descriptions each).
- Utilized computational resources provided by the NAIRR Pilot, including NVIDIA A100 GPUs, to train large-scale AI models for video description tasks.
- Conducted experiments on high-performance computing systems to refine model scalability and performance under real-world constraints.

### Graduate Research Project, San Francisco State University

Jun 2024 - Dec 2025

*“Evaluation of Vision Language Models Using Item Response Theory”*

- Designed and implemented an Item Response Theory (IRT) based evaluation framework to analyze how reliably VLMs apply human rating scales across multimodal tasks.
- Conducted three case studies, image caption quality assessment, comic reading-comprehension inference, and video audio-description evaluation to test model alignment with expert human judgments.
- Applied Partial Credit Models (PCM) to estimate latent rater ability, item difficulty, and threshold structure, enabling direct comparison between human raters and model predictions.
- Validated model performance along with the human agreement using Kendall’s Tau correlation; GPT-4o snapshots achieved alignment scores approaching.

## Work/Teaching Experience

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### TASC Tutor, SFSU

Jan 2023 - Dec 2025

*Tutoring and Academic Support, SFSU*

- Provided one-on-one tutoring for mathematics and computer science courses, including GVAR math proofs, linear algebra, number theory, discrete math, and algorithm classes, tailoring sessions to individual student needs.
- Developed personalized study strategies and clarified challenging concepts, helping students achieve improved academic performance and confidence in technical subjects.

### Software Engineering Intern

Aug 2025 - Nov 2025

*Astro Information Security*

- Developed an email spam detection and a notification system using Gmail API and MCP, enabling real-time classification of incoming messages.

### Teaching Assistant Grader, Number Theory

Aug 2023 - Spring 2025

*Department of Mathematics, SFSU*

- Evaluated and provided constructive feedback on homework assignments, emphasizing clarity and mathematical rigor.
- Responded to student inquiries regarding grading, clarifying misunderstandings and reinforcing course concepts.

### Teaching Assistant Grader, Linear Algebra

Spring 2024 - Spring 2025

*Department of Mathematics, SFSU*

- Reviewed and graded weekly assignments, focusing on the accuracy of calculations and proofs.
- Collaborated with the instructor to develop grading rubrics for assignments.

## Publication/Manuscript

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- Juvenal Francisco Barajas\*, **Gio Jung\***, Manali Seth, Lana Do, Malleeswari Jagabattuni, Andrew Scott, Shasta Ihorn, Vassilis Athitsos, and Ilmi Yoon. *Beyond Lexical Overlap: Validating Semantic Similarity Metrics for LLM-Based Video Caption Augmentation*. In press at ICIIT 2026.
- Alexander Mario Blum, James M. Mason, Robin C. Irely, Yukie Toyama, Yunting Liu, **Gio Jung**, Andrew Scott, JinHo Kim, David Pearson. *Using Item-Response Theory to Disentangle Local vs. Global Processing Dispositions in Autistic Cognition: Comparing Multi-Modal Comics vs. Text-Based Narratives in a Randomized Study*. Journal of Discourse Processes.
- Lana Do, Shasta Ihorn, Charity M Pitcher-Cooper, Sanjay Mirani, **Gio Jung**, Hyunjoo Shim, Zhenzhen Qin, Kien T. Nguyen Mr, Vassilis Athitsos, Ilmi Yoon. *Shaping Human-AI Collaboration: AI Draft Quality Thresholds for Audio Description*. In Review at UIST 2026.
- Lana Do, **Gio Jung**, Juvenal Francisco Barajas, Andrew Scott, Shasta Ihorn, Vassilis Athitsos, Alexander Mario Blum, and Ilmi Yoon. *How well can VLMs rate audio descriptions: A multi-dimensional quantitative assessment framework*. Manuscript at CHI 2026.
- Lana Do, Shasta Ihorn, Charity Pitcher-Cooper, Juvenal Francisco Barajas, **Gio Jung**, Xuan Duy Anh Nguyen, Sanjay Mirani, Ilmi Yoon. *ADx3: A Collaborative Workflow for High-Quality Accessible Audio Description*. Manuscript at CHI 2026.

## Honors/Awards/Grants

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- NAIRR Pilot Resource Oct 2024 - Dec 2025
  - Title: “Scaling AI Video Description Model Capacity for the Blind and Low Vision Community”
- National Science Foundation (NSF), the grant of 3K Jun 2025 - Aug 2025
- Google CAHSI IRP Grant Research Assistant Aug 2024 - Aug 2025
  - Title: “Responsible Design and Development of a Validated AI Video Description Rater”
- Dean’s List, San Francisco State University Spring 2022 - Fall 2023
- Jessie F. L. York Scholarship, 1K Oct 2017

## Poster Presentations

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- Lana Do, Shasta Ihorn, Charity Pitcher-Cooper, Juvenal Francisco Barajas, **Gio Jung**, Xuan Duy Anh Nguyen, Sanjay Mirani, Ilmi Yoon. 2025. *ADx3: A Collaborative Workflow for High-Quality Accessible Audio Description*. NeurIPS 2025. Poster Presentation, December 2025
- **Gio, Jung**, Ilmi Yoon. 2024. *GPT-4o vs. Humans in Rating Image Descriptions*. [Poster Abstract](#). Poster Presentation at CAHSI Research Poster Competition from GMiS Conference, November 2024

## Project

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- SpaceTimeGPT** ([Link](#)) 2025
  - Contributed developing SpaceTimeGPT, a distributed video captioning system integrating TimeS-former and GPT-2 using DeepSpeed Pipeline Parallelism to enable large-scale training on NVIDIA H100 clusters.
- eBay 2025 University Machine Learning Competition** ([Link](#)) 2025
  - Ranked in the top 4 with an F1 score of 0.94 as part of the team “init to win it”.
- ChillMate – AI Mental Health Chatbot** ([Link](#)) Fall 2024
  - Built a chatbot using RAG tailored to SFSU Community; Led Scrum process as a Scrum Master.
- Deep Learning Group Project** Fall 2024
- Data Visualization Group Project** Spring 2024

## Technical Skills

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- Programming languages: Python, SQL, R, pandas, NumPy; data curation, batch inference, CSV/JSON pipelines
- Focus Areas: Artificial Intelligence (AI), Machine Learning, Deep Learning, Computer Vision, Item Response Theory (IRT), Natural Language Processing (NLP), Data Analysis, Statistics, Model evaluation, Mathematics
- ML & AI: PyTorch, Hugging Face Transformers, multimodal models, LLM/VLM evaluation, supervised learning, fine-tuning, prompt engineering
- Other: VS Code, IntelliJ, Jupyter Notebook, Pytorch, TensorBoard, Linux, MacOS, Weights & Biases

## Languages

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**Korean:** Native  
**English:** Fluent